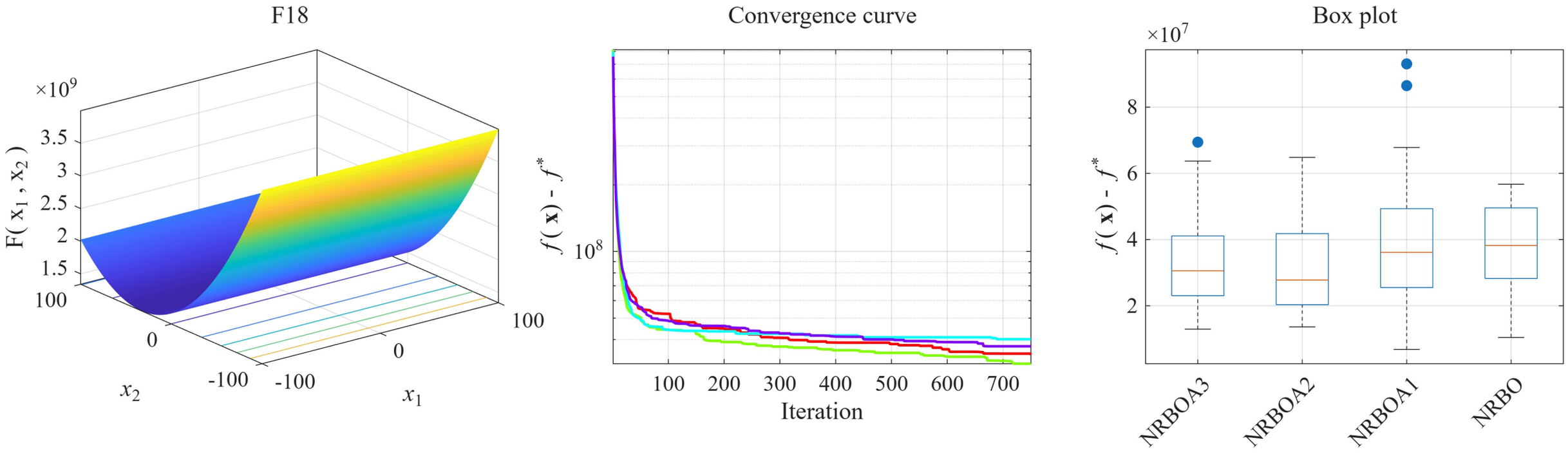

F18
×10^9
3.5
3
2.5
2
1.5
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
10^8
100
200
300
400
500
600
700
Iteration
Box plot
×10^7
8
6
4
2
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO